

High-temperature thermophysical properties and the thermal interpretation of melt-pool geometry in IN625

Abstract

Geometric calibration makes thermal models useful for laser processing, but the solidification history represented by a fitted melt boundary remains insufficiently constrained. High-temperature material assumptions affect both heat redistribution and storage, obscuring the relation between fusion geometry and cooling. Here, a nonlinear enthalpy model separates these effects for bare-plate IN625 tracks by retaining a length-calibrated heat source while varying conductivity and sensible heat-capacity continuations independently. Matching length leaves a fusion zone $19.54\text{ }\mu\text{m}$ wider and $4.88\text{ }\mu\text{m}$ shallower than the reference means. Conductivity holding extends the pool by $43.21\text{ }\mu\text{m}$, whereas heat-capacity holding shortens it by $6.17\text{ }\mu\text{m}$. Rear solidus displacement accounts for 98.3% of the conductivity-induced extension; the solidus–liquidus separation changes much less. Across two scan speeds, rear phase spans near $80\text{ }\mu\text{m}$ correspond to material passage times of 100.33 and $67.33\text{ }\mu\text{s}$. The comparisons distinguish displacement of the thermal tail from enlargement of the phase interval and separate spatial geometry from material cooling time. They establish a controlled basis for interpreting property sensitivity after geometric calibration, while identifying the independent thermal constraints needed to assess the resulting histories.

Keywords: Laser powder bed fusion, IN625, Melt-pool geometry, Enthalpy model, High-temperature properties, Model discrepancy

1. Introduction

Thermal modeling links laser processing conditions to the melting and cooling histories that govern solidification in laser powder bed fusion (LPBF). The required information extends beyond the boundary of a melted track: temperature gradients and material cooling paths determine the thermal conditions presented to a microstructure model. Resolving these fields at practical build scales remains demanding, particularly when laser coupling, phase change and liquid transport are represented together [1, 2]. Melt-pool geometry offers an accessible constraint on less costly calculations, whereas thermography supplies spatial profiles and temporal information [3, 4]. The continuing difficulty is to establish the thermal meaning of a geometric fit when heat-source and material assumptions act together on the calculated field.

Different modeling approaches resolve different parts of this difficulty. Thermo-fluid calculations describe recoil pressure, surface-tension-driven flow and evaporation, including their redistribution of energy within the melt pool [5]. Moving-source conduction offers a simpler description. Rosenthal’s solutions established relations between source motion, heat flow and cooling, while finite-source and nonlinear formulations incorporated distributed heating and temperature-dependent enthalpy [6, 7]. Energy and diffusion-time scaling further reduces the cost of estimating dimensions [8, 9]. Rapid transient conduction calculations have also been assessed against solidification conditions and grain structure for specific laser and electron-beam powder-bed cases [10]. Such simplifications are useful when their assessment addresses the observable and operating regime for which they are applied.

Substantial geometric agreement has been achieved within this hierarchy. Analytical and finite-element calculations reasonably estimated IN718 melt-pool shapes, although their absorption sensi-

tivities and high-input responses differed [11]. In the AMMT IN625 benchmark, a nonlinear phase-change model combined measured laser input with calibrated effective directional conductivity to reproduce the pool dimensions [12]. Source calibration provides a complementary route: inverse boundary-temperature fitting supports interpolation of double-ellipsoid parameters across process conditions [13], while Bayesian treatments of an adapted Eagar–Tsai model propagate geometric uncertainty associated with apparent absorption and powder properties [14]. For bare-plate IN625, a dynamic volumetric source calibrated over multiple beam sizes was subsequently applied to a layer-scale benchmark [15]. These results demonstrate that source representation and effective transport are substantive components of geometric prediction, rather than interchangeable fitting conventions.

Their implications for temperature require a separate assessment. Myers et al. obtained similar geometric fits from parameter combinations that produced different surface-temperature profiles; two-color imaging provided an additional constraint [4]. Effective energy coupling is itself condition-dependent. Scanning calorimetry relates coupling and penetration to power, speed and beam size across the conduction–keyhole transition [16], while IN625 measurements distinguish unreflected laser power from energy delivered to the substrate [17]. Observation definitions introduce a further distinction: AMMT imaging length and metallographic width/depth are obtained through different procedures and sampled track populations [18]. A fitted source factor therefore reflects the model and observation used to determine it; agreement in the fitted dimension does not independently constrain the remaining shape or cooling path.

Recent studies clarify the need for this output-specific interpretation. Gaussian conduction comparisons with thermal imaging and metallography found useful surface trends together with depth bias, under constant-property and phase-change assumptions different from a nonlinear enthalpy model [19]. Analytical sensitivity studies quantify how processing and material inputs affect geometry and temperature [20]. A separate bare-track failure test inverts measured depth to determine whether a conduction model would require physically infeasible absorption [21]. These established approaches address reliability, parameter response and model feasibility. The unresolved issue addressed here is narrower: after a geometric calibration, the thermal significance of a high-temperature property change cannot be read directly from its effect on overall pool length.

This issue arises when input tables end below the phase-change interval. The IN625 conductivity and sensible heat-capacity data used here terminate below the adopted solidus [22]. Their continuation determines both heat redistribution and enthalpy storage beyond the tabulated range. In a phase-change formulation, a change in sensible heat capacity is distinct from a change in latent-inclusive storage. Likewise, a displacement of the rear solidus and liquidus together can extend the pool without comparably widening the interval between them. Material time introduces another distinction: passage through a translating temperature field depends on scan speed as well as spatial phase separation. These relations motivate a comparison that resolves individual property effects and phase-boundary motion rather than treating total length as a surrogate for the cooling history.

The present study uses a nonlinear moving-frame enthalpy model for three AMMT bare-plate IN625 conditions. One effective source factor is calibrated to surface length and retained for comparisons with the excluded width/depth measurements and the other operating conditions. A fixed-source factorial comparison separates linear continuation and last-value holding of conductivity and sensible heat capacity. The resulting geometry changes are decomposed into rear solidus/liquidus displacement and phase-span change, then related to material passage time and interval-mean cooling rate. The analysis connects thermophysical-property sensitivity to specific features of the modeled thermal history while preserving the distinction between geometric comparison, numerical verification and independent thermal validation.

2. Materials and methods

2.1. Benchmark conditions and comparison design

The study examines whether matching one melt-pool length also constrains fusion shape and the rear thermal interval in a conduction model. One source factor is fitted to condition B length. Width and depth are excluded from the fit, and conditions A and C retain the fitted factor. A four-corner comparison at B then changes the high-temperature conductivity and sensible-storage continuations independently. The rear solidus and liquidus positions resolve whether these changes displace the thermal tail or widen the phase interval; their separation and scan speed define a material passage time.

The experimental comparison uses constant-speed laser scanning on bare IN625 plate under three public AMMT conditions (Table 1). These are single tracks without a powder layer. The three conditions cover increasing speed; A also has a different power, whereas B and C isolate a speed change at fixed nominal power. The measured beam diameter is $D_{4\sigma} = 170 \mu\text{m}$ [18]. For the normalized circular Gaussian used here, $D_{4\sigma} = 2w$, giving $w = 85 \mu\text{m}$; it does not reproduce the measured two-dimensional beam map.

Table 1: Operating conditions and fixed-source energy scales. The effective factor $\eta \approx 0.28905$ is fitted to B length only. $E_\ell = \eta P/v$ and w/v are model input scales.

Condition	P (W)	v (m/s)	E_ℓ (J/m)	w/v (μs)
A	137.9	0.4	99.65	212.50
B	179.2	0.8	64.75	106.25
C	179.2	1.2	43.17	70.83

The length reference is the AMMT condition mean reported by Lane et al. [18]. Width and depth means are taken from NIST challenge Table 2 for the 100 μs -track microscopy classes [23]. The latter differ in aggregation and display precision from the subsequently published cross-section means in Lane et al., which combine 100 and 20 μs tracks. Length and cross-section references are therefore condition-level quantities from separate observation populations, rather than a joint three-dimensional measurement of one track.

The comparisons are retrospective and nonblind: all public targets were available during development, although only B length enters the calibration objective. Expanded uncertainties U with coverage factor $k = 2$ are taken separately from Lane’s uncertainty budgets. For width and depth these are descriptive measurement scales from the published combined-class analysis, not uncertainty values jointly estimated with the current decimal NIST means. The ratio δ/U , where δ is the signed model–reference difference, is used only to report discrepancy scale. It is not a significance test or a combined numerical, input, calibration and measurement uncertainty statistic.

2.2. Moving-frame enthalpy model and material continuations

The laboratory-frame conduction balance uses conservative enthalpy [7]:

$$\frac{\partial H}{\partial t} = \nabla \cdot [k(T)\nabla T], \quad H(T) = \rho \left[\int_{T_0}^T c_p(s) \, ds + \mathcal{L}f(T) \right]. \quad (1)$$

The density $\rho = 8440 \text{ kg m}^{-3}$ and melting limits $T_s = 1290 \text{ }^\circ\text{C}$ and $T_l = 1350 \text{ }^\circ\text{C}$ follow the IN625 bulletin [22]. The latent heat $\mathcal{L} = 280000 \text{ J kg}^{-1}$ follows the earlier AMMT model [12]; the reference

temperature is set to $T_0 = 25$ °C. A linear liquid-fraction closure is adopted over that melting interval, following an established continuum-model approximation [15]

$$f(T) = \begin{cases} 0, & T < T_s, \\ (T - T_s)/(T_l - T_s), & T_s \leq T < T_l, \\ 1, & T \geq T_l. \end{cases} \quad (2)$$

Specific heat and conductivity are linearly interpolated between the input knots in Table A.4. The conductivity table contains 15 points and ends at 982 °C; the 12-point specific-heat table ends at 1093 °C. The supplier labels its typical specific heats as calculated and conductivity as measured [22]; these are general alloy data, not measurements on the benchmark coupon. Differentiating Eq. (1) within the adopted phase interval gives $c_{p,\text{eff}} = c_p + \mathcal{L}/(T_l - T_s)$, with a latent contribution of $4666.67 \text{ J kg}^{-1} \text{ K}^{-1}$.

Both property tables end below the adopted solidus, so the melting-range solution requires a high-temperature continuation (Fig. 1).

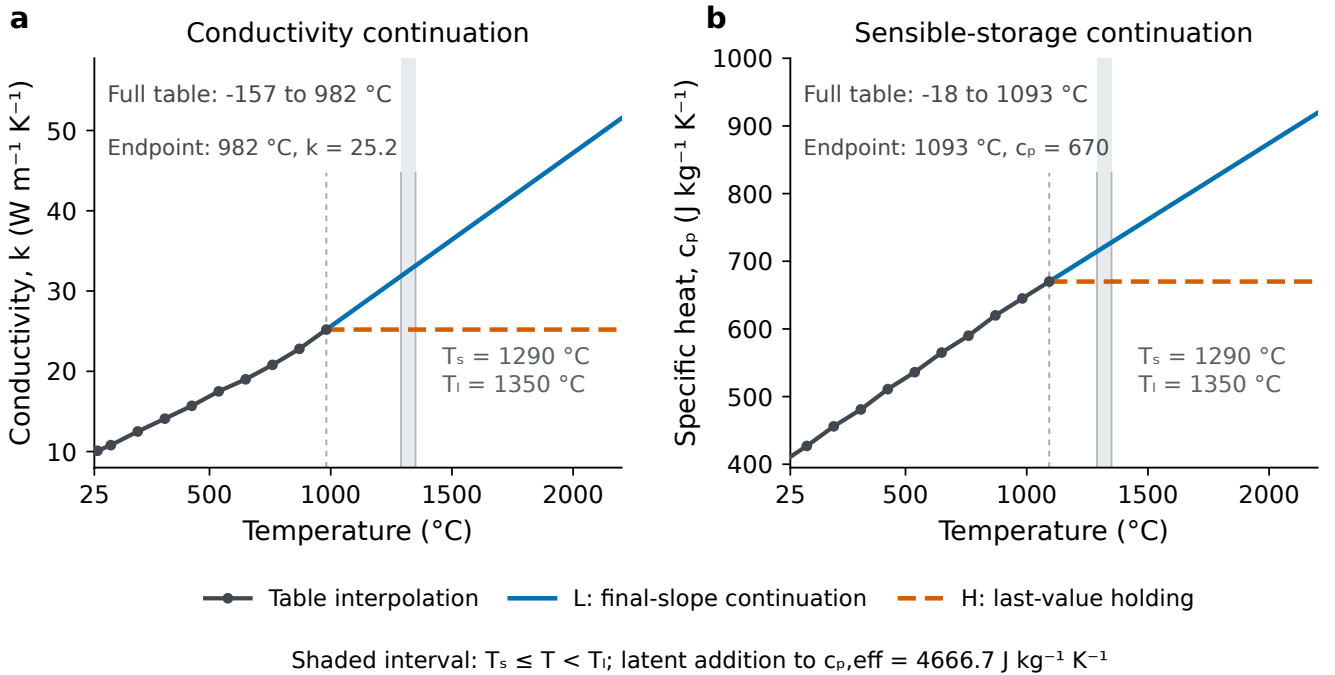


Figure 1: Actual IN625 property inputs and the two continuation choices. (a) Conductivity and (b) sensible specific heat use the input knots listed in Table A.4, with common piecewise-linear interpolation below the respective endpoints. The final knots are 982 °C and $25.2 \text{ W m}^{-1} \text{K}^{-1}$ for k , and 1093 °C and $670 \text{ J kg}^{-1} \text{K}^{-1}$ for c_p . L continues the final tabulated secant; H holds the final value. The displayed window is 25–2200 °C; all 15 conductivity and 12 specific-heat source rows, including those below 25 °C, are retained in the accompanying data and used in interpolation. Shading marks the adopted 1290–1350 °C phase interval. The latent addition to effective specific heat within this interval is $4666.7 \text{ J kg}^{-1} \text{K}^{-1}$ and is stated separately from sensible c_p . The supplier describes conductivity as measured and specific heat as calculated typical alloy data [22]; the plotted high-temperature continuations are modeling choices.

Two binary factors replace linear continuation (L) by holding the last tabulated value (H), independently for k and c_p . H holds $k = 25.2 \text{ W m}^{-1} \text{K}^{-1}$ above 982 °C and $c_p = 670 \text{ J kg}^{-1} \text{K}^{-1}$ above 1093 °C. L extends the final tabulated secant of each property; the appendix lists the corresponding slopes. The changes apply above each table endpoint, including the solid range between the endpoint and T_s . Each conductivity branch changes k , its primitive K and the surface inverse together; each

storage branch changes c_p , H and $T(H)$ together. The first letter labels conductivity. The baseline uses linear continuation for both properties (LL); HL and LH change one property, and HH changes both. The calibration and fixed-source contrasts are defined below.

Following the moving-source coordinate description [7], let $\xi = x - vt$. For a quasi-steady field, $\partial_t H = -v \partial_\xi H$; defining $\mathbf{u} = (-v, 0, 0)$ therefore gives

$$\nabla \cdot (\mathbf{u}H - k \nabla T) = 0. \quad (3)$$

The term $\mathbf{u}H$ represents stationary material passing through the laser coordinate frame. No liquid momentum equation or melt convection is included. The assumptions also exclude evaporation and free-surface deformation.

With the circular Gaussian $D_{4\sigma}$ convention [15], the surface input at $z = 0$ is

$$q(\xi, y) = \frac{2\eta P}{\pi w^2} \exp\left[-\frac{2(\xi^2 + y^2)}{w^2}\right]. \quad (4)$$

With outward normal $\mathbf{n}$, the top condition is $-k \nabla T \cdot \mathbf{n} = -q$; the Gaussian supplies inward heat. Its full-plane integral is ηP ; the appendix gives the discrete source integration. At positive ξ , incoming material has $H = 0$; at negative ξ , diffusive gradient vanishes and enthalpy leaves advectively. The $y = 0$ plane is symmetric. Outer transverse and bottom boundaries are adiabatic. Radiation and ambient convection are omitted from the solved balance.

2.3. Melt-pool geometry and material passage

Model length L is the distance between two solidus crossings of the recovered surface symmetry-centerline temperature. Crossings are interpolated, must enclose the source, and must form one interval without domain truncation. Surface and symmetry-plane recovery precede threshold extraction; the appendix gives their numerical definitions.

Fusion-zone width and depth are extracted from the source-connected component of

$$T_{\max}(y, z) = \max_{\xi} T(\xi, y, z) \geq T_s. \quad (5)$$

Each ξ slice is first reconstructed at the surface and symmetry plane, then bilinearly interpolated onto observation coordinates, and only then maximized over passage. Taking a nodal maximum before interpolation creates a different envelope. Width is twice the positive- y extent and depth is the negative of the lowest z . The observation pitch samples interpolated geometry and does not define the heat-field resolution. Figures label the surface length proxy L_s and fusion-envelope extents W_f and D_f ; these correspond to L , W and D in the text.

Lane's imaging length uses a radiance-profile second-derivative minimum for the rear solidification location and a calibrated front threshold [18]. The present two-solidus-crossing proxy does not replay radiance, emittance, pixel integration, viewing geometry or motion blur. Its fitted η is therefore an effective source/model/operator parameter. Width and depth refer instead to a maximum-temperature fusion envelope, compared with metallography.

Let ξ_s^- and ξ_l^- denote the rear centerline solidus and liquidus crossings. Define the rear spatial phase span, passage time and interval-mean cooling-rate magnitude as

$$\ell_m = \xi_l^- - \xi_s^-, \quad \tau_m = \frac{\ell_m}{v}, \quad \bar{T} = \frac{T_l - T_s}{\tau_m}. \quad (6)$$

These quantities describe stationary material passing through the same quasi-steady field; they are derived quantities rather than independent thermal observations. The conversion follows from $d\xi/dt = -v$ for a fixed laboratory point. Constant-speed space-to-time mapping is also used in the AMMT measurements [18], whose below-solidus cooling interval differs from the present 1350–1290 °C interval.

2.4. Calibration, constitutive contrasts and numerical adequacy

Fitting $L_B(\eta) = 359 \mu\text{m}$ gives $\eta \approx 0.28905$, with a signed length residual of $+0.158 \mu\text{m}$. Conditions A and C retain this value, beam, phase law and property continuations. The appendix gives the scalar calibration algorithm, trial values and local sensitivity.

For the four B corners, power, speed, beam, density, phase thresholds, latent heat, mesh and observation operators remain fixed, and no corner is recalibrated. The first letter denotes conductivity and the second sensible specific heat. Sharing B=LL between the operating-condition and continuation comparisons gives six production solutions.

For any output Q , define the baseline-anchored finite effects

$$\delta_k = Q_{\text{HL}} - Q_{\text{LL}}, \quad \delta_{c_p} = Q_{\text{LH}} - Q_{\text{LL}}, \quad (7)$$

$$I_{k,c_p} = Q_{\text{HH}} - Q_{\text{HL}} - Q_{\text{LH}} + Q_{\text{LL}}, \quad \Delta_{\text{joint}} = Q_{\text{HH}} - Q_{\text{LL}} = \delta_k + \delta_{c_p} + I_{k,c_p}. \quad (8)$$

The conditional conductivity effect at held c_p is $Q_{\text{HH}} - Q_{\text{LH}} = \delta_k + I_{k,c_p}$; the conditional storage effect at held k is $Q_{\text{HH}} - Q_{\text{HL}} = \delta_{c_p} + I_{k,c_p}$. Reporting both conditions checks whether a direction changes with the other factor. These deterministic contrasts distinguish two continuation choices within the same model. They are not estimates from replicated material experiments and have no sampling-significance interpretation.

An axial-face Péclet diagnostic compares coordinate enthalpy transport with diffusion in the phase interval. A discrete rear hot-region budget measures net outward diffusive power. Both use temperature-selected regions that can change between branches, so they complement matched-coordinate thresholds rather than supply a fixed-region causal contrast. Their definitions are given in the appendix.

The conservative finite-volume solution uses near-source cell widths of $4 \times 3 \times 1 \mu\text{m}$. The graded-domain geometry and enthalpy discretization are specified in the appendix. Convergence is checked using both the sum of absolute cell residuals and the signed global energy balance, followed by a full frozen-coefficient correction.

A separate constant-property moving-Gaussian problem checks source integration, frame transport and geometric extraction against a semi-infinite analytical reference. Verification here tests numerical implementation separately from physical validation [24]. Its reference $L/W/D$ is $407.535/153.293/34.836 \mu\text{m}$; finite-volume differences are $+0.139\%$, $+0.190\%$ and -0.499% , respectively. This simpler comparison does not establish nonlinear phase-change accuracy.

At the calibrated η , reducing the near-source axial spacing from 4 to $2 \mu\text{m}$ changes B length, width and depth by $+0.361$, -0.009 and $+0.041 \mu\text{m}$, and C by $+0.199$, -0.029 and $+0.045 \mu\text{m}$. A/B domain enlargement changes these quantities by less than $3 \times 10^{-6} \mu\text{m}$. These measured sensitivities support comparison with the larger geometric responses, but do not give a full three-dimensional error bound for each continuation branch. Submicrometre interactions are therefore reported descriptively. An earlier transient B comparison at a different calibration stage provides only approximate local steady consistency. Fixed-field blackbody estimates put B-corner top losses at 0.099 – 0.180% of absorbed power; feedback from these losses is not solved.

All six solutions satisfy the stated energy-balance and nonlinear-correction criteria and give bounded phase fractions and nontruncated extracted geometry. The appendix reports the acceptance thresholds and final residuals.

3. Results

The length-calibrated model is first compared with the remaining geometric measurements. The property contrast then separates changes in overall shape from motion of the rear phase boundaries, before the scan-speed comparison relates that spatial structure to material time.

3.1. Melt-pool geometry across scanning conditions

The one-target calibration reproduces the B length proxy while leaving a systematic width/depth residual at B and C (Table 2 and Fig. 2). The fitted length is $359.16 \mu\text{m}$, but the B width exceeds the nominal reference by $19.54 \mu\text{m}$ and depth falls below it by $4.88 \mu\text{m}$. Their separate expanded-uncertainty-scale ratios are $+3.10$ and -1.23 . Thus length fitting does not reproduce the remaining B shape quantities.

Table 2: Condition-level geometry comparisons in μm . U is the separately sourced Lane expanded uncertainty ($k = 2$). Reference width/depth means are current NIST 100 μs -track classes. B length is fitted. δ/U uses only the experimental scale, without combined model uncertainty.

Condition	Quantity	Model	Reference	U	δ/U
A	L	309.89	300	11.91	+0.83
	W	153.36	147.9	6.42	+0.85
	D	40.47	42.5	4.49	-0.45
B	L (fitted)	359.16	359	21.26	+0.01
	W	143.04	123.5	6.30	+3.10
	D	31.12	36	3.97	-1.23
C	L	320.63	370	32.57	-1.52
	W	129.05	106	4.48	+5.14
	D	22.04	29.6	3.26	-2.32

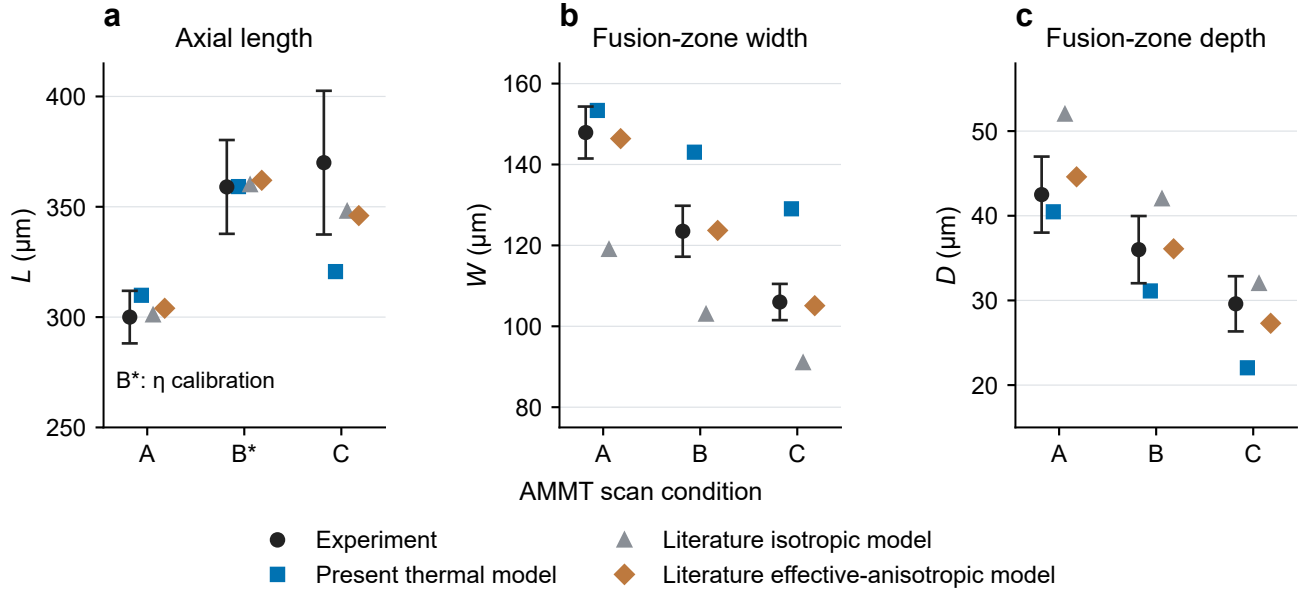


Figure 2: Geometry comparison with condition-level reference means and the published isotropic and effective-anisotropic calculations of Kollmannsberger et al. [12]. Experimental bars show separately sourced Lane $U(k = 2)$ scales; width/depth centers use current NIST means. B length is calibrated in the present model. Published-model points provide context: beam representation, phase law, parameter calibration and observation definitions differ, so this is not a controlled ranking of solvers.

A differences are smaller than their respective reported uncertainty scales. In C, the model is $49.37 \mu\text{m}$ shorter, $23.05 \mu\text{m}$ wider and $7.56 \mu\text{m}$ shallower than the nominal references. The wide/shallow pattern is also apparent in the predicted aspect ratios $W/D = 3.79, 4.60, 5.85$ for

A/B/C, compared with condition-level reference ratios 3.48,3.43,3.58. These ratios combine separate class means; they are not specimen-wise aspect-ratio measurements.

At fixed power from B to C, a 50% speed increase reduces absorbed line energy by one third. Model width decreases by 9.78% and depth by 29.16%; the nominal reference changes are 14.17% and 17.78%. The modeled W/D increases by 27.36% from B to C. The shape discrepancy therefore concerns the relative contraction of transverse and depth extents as well as their absolute values. A–B changes cannot be attributed to speed alone because power also changes.

Nominal experimental length increases by 11 μm from B to C, whereas model length decreases by 38.53 μm . The 11 μm increase is smaller than the 38.89 μm zero-correlation quadrature of the two reported expanded-uncertainty scales. The nominal increase is therefore not resolved by this uncertainty-scale comparison. C’s individual discrepancy remains distinct from that trend comparison. Retained axial changes below 0.37 μm are far smaller than the main B/C residuals; refining that direction alone is unlikely to remove the observed shape pattern.

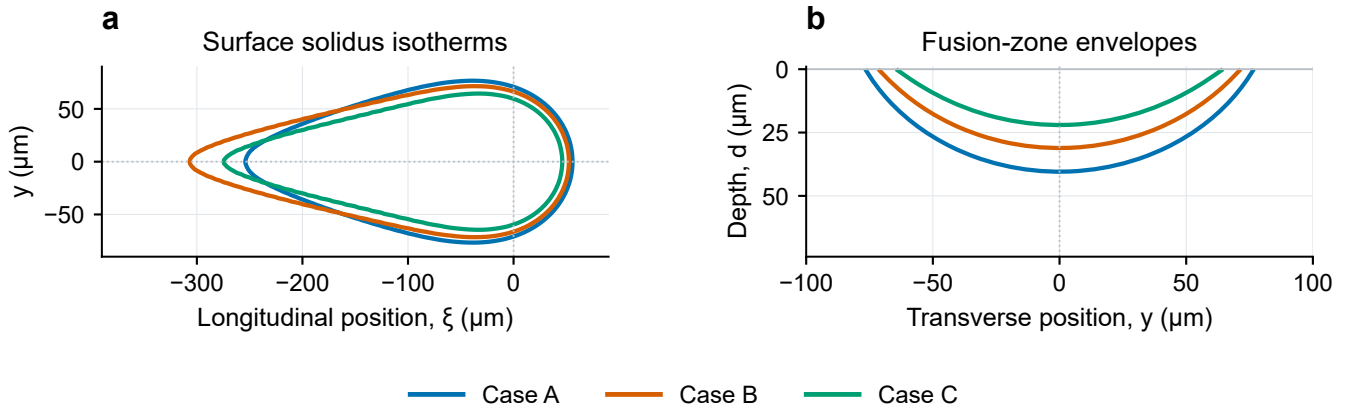


Figure 3: Model geometry for A, B and C. (a) Surface solidus contours in laser-frame coordinates. (b) Reconstructed maximum-temperature fusion envelopes in transverse/depth coordinates. Fusion envelopes use slice reconstruction and interpolation before passage maximization. The fields are model outputs, not microscopy or inferred liquid-flow maps. Both panels use the fixed calibrated source factor.

The contours in Fig. 3 show the preferential reduction in penetration for the fixed-power B–C comparison. The surface solidus contour determines the length proxy, while the transverse envelope records the maximum melted extent during material passage. Their differing responses motivate the property comparison at fixed source conditions.

3.2. High-temperature property effects on the thermal tail

The four-corner results are shown in Table 3 and Fig. 4. Holding conductivity at its last table value increases length from 359.16 to 402.37 μm at linearly continued c_p . Holding c_p instead decreases length to 352.99 μm . Holding both yields 393.44 μm . The joint increase is therefore

$$\Delta L_{\text{joint}} = 43.21 - 6.17 - 2.76 = 34.28 \mu\text{m}. \quad (9)$$

The heat-capacity response and interaction offset part of the conductivity response. A comparison of LL and HH alone would miss this decomposition. The signs persist at the other factor level: conductivity holding increases length by 40.45 μm when c_p is held, and storage holding decreases length by 8.93 μm when k is held. Conductivity therefore extends the tail and storage holding shortens it in both tested settings, with the interaction changing the magnitudes.

Table 3: Fixed-B deterministic continuation contrast. The first letter labels k , the second c_p ; L means linear continuation and H last-value holding above the respective endpoints. No branch is recalibrated. Rounded geometry and effects are in μm . Small transverse effects lack branch-specific refinement evidence.

Corner or contrast	L	W	D
LL	359.16	143.04	31.12
HL	402.37	143.15	30.09
LH	352.99	143.72	31.61
HH	393.44	143.79	30.80
δ_k	+43.21	+0.11	-1.03
δ_{c_p}	-6.17	+0.68	+0.49
I_{k,c_p}	-2.76	-0.04	+0.21
Δ_{joint}	+34.28	+0.75	-0.32

Conductivity holding moves the rear solidus and liquidus by -42.49 and -43.50 μm , while the front solidus moves by $+0.73$ μm . The rear solidus displacement accounts for 98.3% of the conductivity-induced length increase; the front contributes the remaining 1.7%. The rear phase span changes by only -1.01 μm . Heat-capacity holding moves the rear positions forward by $+6.22$ and $+6.53$ μm , changing their separation by $+0.31$ μm . The large length response thus predominantly translates both rear phase boundaries together. It does not represent comparable broadening of the mushy interval.

At 1320 °C, conductivity changes from 32.51 to 25.20 $\text{W m}^{-1} \text{K}^{-1}$ under holding, a reduction of 22.48%. Sensible specific heat decreases from 721.13 to 670 $\text{J kg}^{-1} \text{K}^{-1}$, or 7.09%. With the latent contribution included, effective specific heat decreases from 5387.79 to 5336.67 $\text{J kg}^{-1} \text{K}^{-1}$, or 0.95%, and enthalpy from the cold reference decreases by 0.66%. This matched-temperature comparison describes the adopted mushy interval. A retained local face Péclet diagnostic for the mushy ξ faces increases in median from 4.44 in LL to 5.77 in HL, consistent with reduced diffusion relative to coordinate enthalpy transport. The diagnostic is consistent with a conductivity-dependent shift in the model’s transport balance. The corresponding retained rear hot-region outward diffusive flux rises from 14.724 to 14.913 W. Since that region is selected by temperature and changes with the solved state, conductivity holding cannot be equated with a measured reduction in its integrated outward heat flow. The branch also changes surface recovery consistently with k .

The current-mesh depth contrast illustrates cancellation. Conductivity holding changes it by -1.03 μm ; heat-capacity holding contributes $+0.49$ μm and the interaction $+0.21$ μm , leaving a net -0.32 μm . All corners remain wider and shallower than the B reference. The conductivity depth effects remain negative at both storage levels (-1.03 and -0.82 μm), whereas the storage depth effects remain positive at both conductivity levels ($+0.49$ and $+0.71$ μm). These two continuation options therefore do not repair the baseline shape discrepancy. The main 34–43 μm length response substantially exceeds the retained baseline axial sensitivity; the 0.04 μm width interaction has no corresponding precision demonstration and is not interpreted as a robust fine ranking.

Across the four corners, the rear phase span lies between 79.25 and 81.17 μm while total length ranges from 352.99 to 402.37 μm . The contrast consequently changes the location of the rear phase interval much more than its spatial separation. This is the main comparison supplied by the retained centerline profiles; the small span ordering remains a current-mesh result.

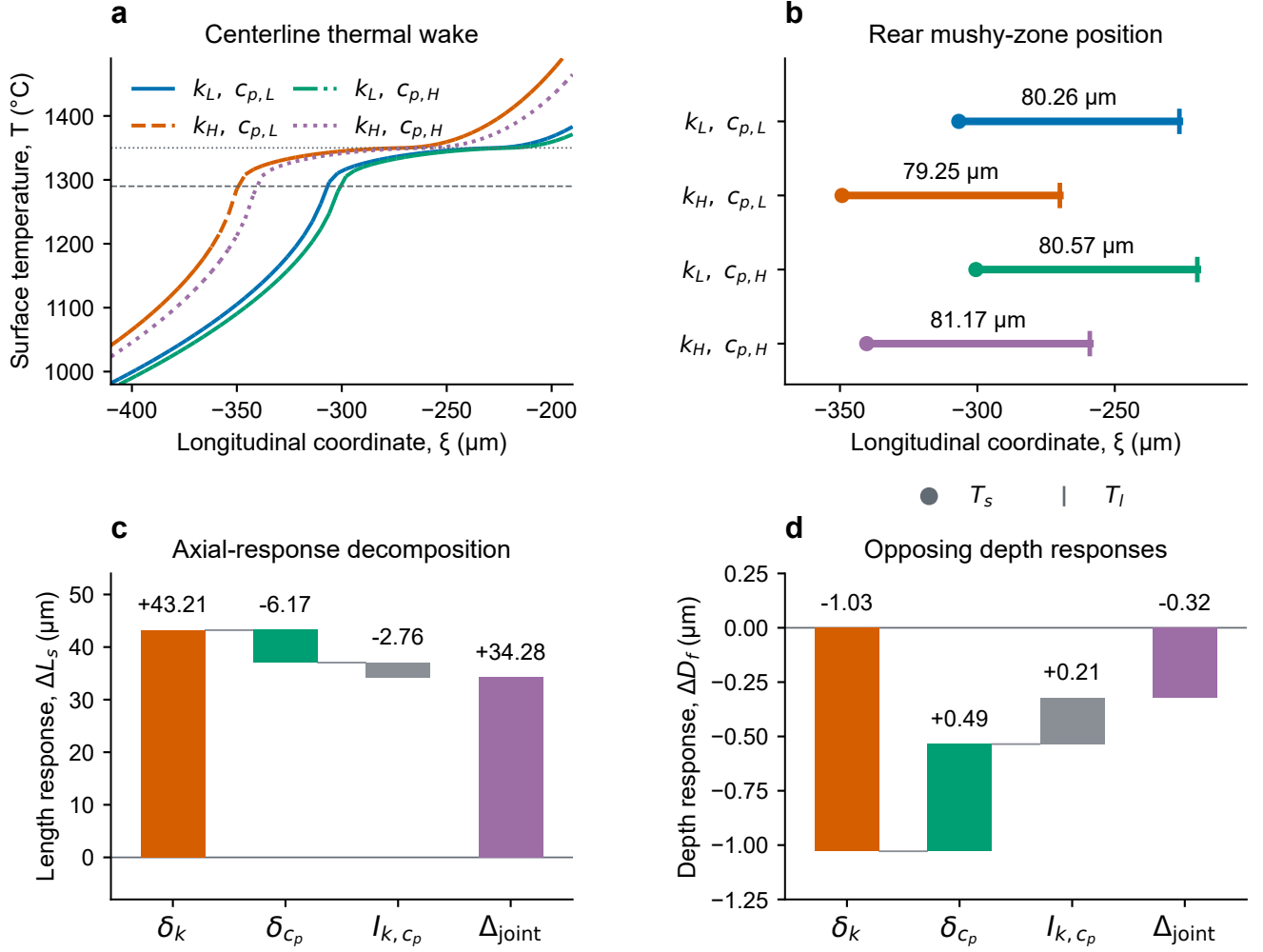


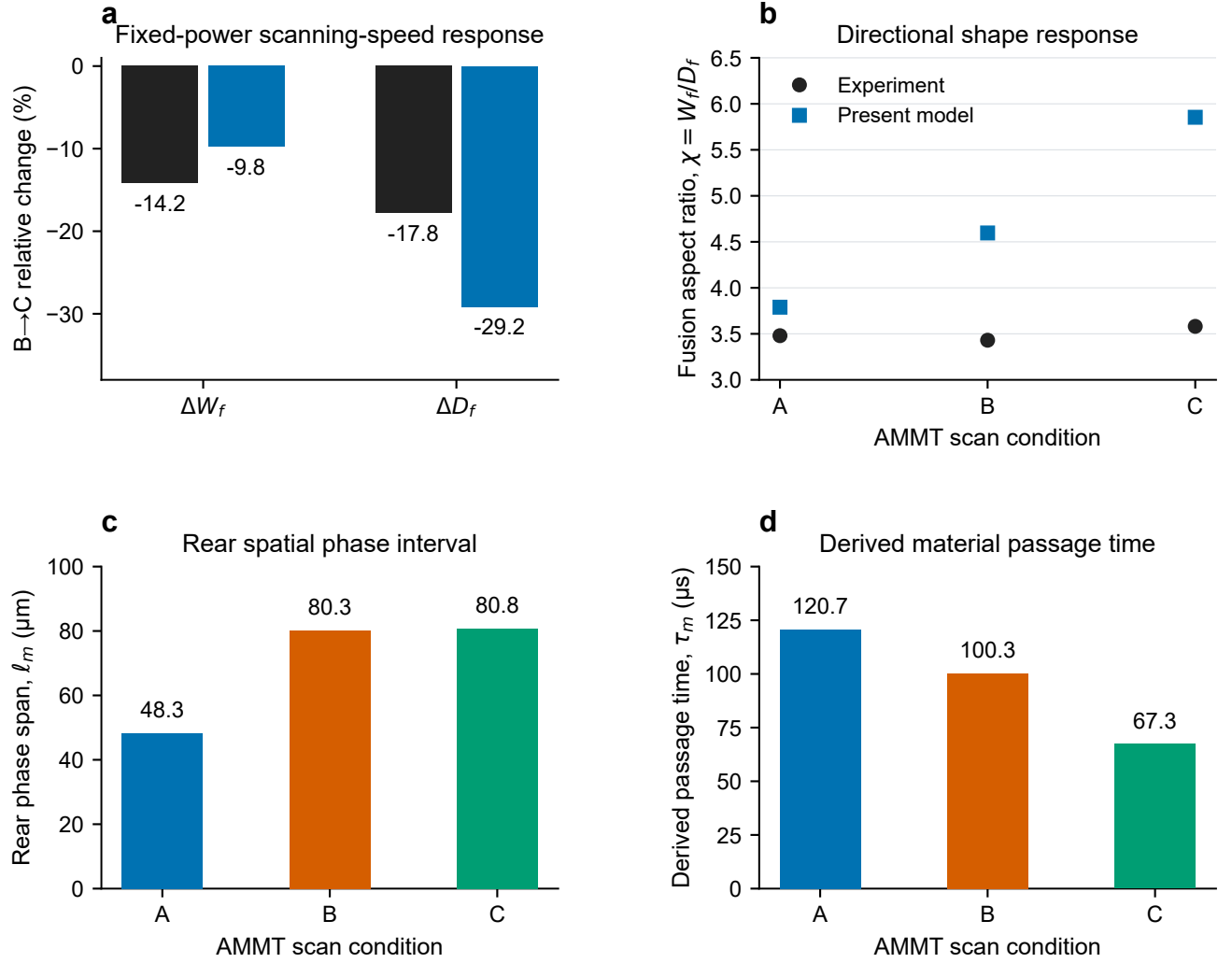
Figure 4: Four-corner continuation comparison at fixed B: rear centerline temperature profiles, phase-boundary positions and spatial span, and baseline-anchored geometry effects. The same phase thresholds, source factor and mesh are retained. Profiles are plotted from original field values with segment clipping to the displayed window. Factorial effects are deterministic finite contrasts, not statistical effect estimates. Large rear translation dominates the length response; joint depth change conceals opposing single-factor contributions.

3.3. Scan-speed dependence of the rear thermal interval

On the present mesh, the rear centerline phase spans are 48.30, 80.26 and 80.79 μm for A/B/C (Fig. 5). The B/C spans differ by only 0.53 μm , yet their transformed passage times are 100.33 and 67.33 μs . The C span is 0.66% larger, while the passage time is 32.90% shorter because speed increases from 0.8 to 1.2 m/s. The corresponding interval-mean cooling rate rises from 0.598 to 0.891 MK s^{-1} ; A gives 120.74 μs and 0.497 MK s^{-1} . Thus the B-to-C interval-mean cooling rate increases by 49.02% despite only a 0.53 μm difference in the spatial phase span.

The 49.02% increase follows from the shorter passage through an almost unchanged spatial phase interval, using Eq. (6). It therefore describes the thermal history along the modeled centerline; its dependence on scan speed cannot be inferred from the spatial span alone.

The operating-condition comparison therefore separates a change in the modeled fusion shape from a change in material passage time. The latter follows from the rear thresholds and speed; its experimental counterpart is not included in the geometric comparisons.



B→C: $P = 179.2 \text{ W}$; relative changes use reported means, with uncertainty not propagated.

Figure 5: Operating-condition response. (a) Nominal B-to-C percentage changes in width and depth at fixed power, comparing model and condition-level reference means; uncertainty is not propagated. (b) Model and reference class-mean aspect ratios. (c) Model rear liquidus-to-solidus spatial spans. (d) Derived material passage times. The model histories share one steady solution and should not be counted as independent validation observables. Interval-mean cooling rates discussed in the text are derived over 1350–1290 °C; they are not plotted here.

4. Discussion

4.1. Geometry calibration and model discrepancy

The length fit constrains one combination of source strength, material continuation and observation mapping. It leaves the B fusion envelope 19.54 μm wider and 4.88 μm shallower than the nominal references, and the fixed-parameter B-to-C shape response differs from the experimental means (Fig. 2). These discrepancies show which information the fit leaves unresolved: penetration, transverse heat distribution and the temperature field are not independently constrained by the surface length. Source representation, transport closure and the distinction between radiance-derived and temperature-threshold geometry remain possible contributors.

Geometric agreement can conceal thermal differences even in a more detailed model. In the bare-plate 316L calculations of Myers et al. [4], several Fresnel/accommodation parameter combinations gave similar agreement with cross-sectional area and width-to-depth ratio but different surface tem-

perature profiles. Two-color measurements narrowed those combinations. The present comparison has a different design: only the LL length is fitted, and the other continuations retain its source factor. Their thermal differences measure response after one calibration; they do not demonstrate thermal differences between equally refitted branches.

The closest AMMT comparison also makes source and transport choices consequential. Kollmannsberger et al. [12] used measured laser input and an effective directional conductivity calibrated at B to improve the A/C geometric comparisons, while explicitly limiting the validity of internal melt-pool temperatures. Their result indicates a route to better geometry under a different source and closure, rather than a ranking against this isotropic Gaussian calculation. The current continuation experiment is useful precisely because it preserves those choices and asks what changes when the missing high-temperature property range is treated differently.

4.2. Conductivity and storage in the thermal tail

Keeping the calibrated source fixed reveals opposing property responses that a joint replacement would conceal. Conductivity holding extends the pool at either storage setting, while sensible-storage holding shortens it at either conductivity setting (Table 3). Their combination produces a 34.28 μm length increase, with the interaction modifying the magnitudes. The rear profiles locate the dominant response: the rear solidus accounts for 98.3% of the conductivity-induced length increase, and the liquidus moves almost the same distance. The principal change is a displacement of the thermal tail relative to the source, rather than comparable widening of the rear phase interval (Fig. 4). None of the four corners removes the B wide/shallow discrepancy.

The effective-heat-capacity representation separates latent and sensible storage [7]. Applied to the adopted linear phase law, it explains why the local storage perturbation is much smaller than the sensible- c_p perturbation. Inside the phase interval, the fixed latent term contributes 4666.67 J kg⁻¹ K⁻¹ to effective specific heat. At 1320 °C, holding sensible c_p reduces it by 7.09%, but reduces latent-inclusive effective specific heat by only 0.95% and enthalpy from the cold reference by 0.66%. The conductivity reduction at the same temperature is 22.48%. These comparisons show that the two nominal property changes perturb transport and total storage by different amounts. They do not isolate latent heat as the cause: latent heat is unchanged in all corners, and each continuation starts below the solidus, so pre-melting enthalpy and the temperature–enthalpy inverse also change.

The diagnostics support a changed balance between coordinate enthalpy transport and diffusion, but do not support equating conductivity holding with reduced integrated outward heat flow from the selected rear region. The selected axial-face Péclet diagnostic increases under conductivity holding, but the integrated rear-region conductive power also increases (Section 3.2). Lower conductivity therefore does not imply lower integrated outward heat flow in this comparison; both the temperature gradients and the boundary of the temperature-selected region change. Surface recovery also changes consistently through $K(T)$ and its inverse. The resolved tail displacement is a response of the complete conduction model and observation operator. Assigning it solely to suppressed bulk heat escape would require a matched-region flux comparison or a separation of field and recovery contributions.

4.3. Spatial phase span and material cooling time

Rear-boundary location, phase separation and material time describe different features of a fitted field. The current-mesh B/C rear phase spans are 80.26 and 80.79 μm , while their passage times are 100.33 and 67.33 μs because scan speed rises from 0.8 to 1.2 m/s (Fig. 5). Thus nearly unchanged spatial separation coexists with a much shorter material interval. The 49.02% increase in interval-mean cooling rate follows from the same thresholds and the division by speed in Eq. (6); it is not

an additional independent thermal test. Likewise, translating the two rear boundaries together can change total pool length substantially without producing a similar change in passage time.

Thermography distinguishes these questions experimentally. Hooper [3] measured both centerline spatial profiles and temperature histories at fixed build-plate locations, including pulsed scans and turn events. Those observations show why a spatial profile and a thermal history cannot be exchanged without accounting for the scan motion. Lane’s cooling-rate exemplars [18] use a below-solidus interval and a different rear-boundary definition, so they do not validate the present 1350–1290 °C interval. A comparison would require a temperature trajectory over the same interval, with scan position, exposure and observation mapping included.

The passage transformation describes stationary material crossing the assumed steady field. Fluid trajectories can sample a different thermal path: in a Ni–20Cr dispersion-strengthened alloy, melt-flow calculations produced narrower near-surface peaks and hotter subsurface excursions than an unmodified conduction field [25]. That material and flow model do not validate the present IN625 interval, but show why ℓ_m/v cannot be equated with liquid residence or a nonequilibrium freezing path. This boundary matters when thermal outputs are used for microstructure: Plotkowski et al. [10] treated melt-pool geometry verification separately from solidification-condition calculations and an experimental grain-structure comparison. Their AlSi10Mg/IN718 study provides a useful methodological distinction, not validation of this IN625 model. Here, neither grain morphology nor segregation has been tested, and the rear passage interval remains a model diagnostic. In transient multi-line calculations, interface velocity can also depart from beam velocity [10]; the present v -based conversion is restricted to a straight, quasi-steady track.

4.4. Thermal interpretation and discriminating evidence

The shape residual does not identify a unique missing mechanism. An alternative source is a substantive explanation: Coleman et al. [15] calibrated a dynamic volumetric source for bare IN625 at 195 W and 800 mm/s over several spot sizes, then applied it to a layer case. Their source adjusts depth and effective absorption with the local melt-pool state; their phase-dependent properties and 1410 K eutectic endpoint also differ from the present equilibrium interval. A comparison using the same measured beam, phase assumptions, calibration target and geometry operators would distinguish source redistribution from the effect of a different material closure. An imaging replay is a cheaper first test of how the radiance-to-threshold mismatch changes the fitted source factor. Neither comparison is supplied by the present continuation corners.

Two physical limits govern the thermal interpretation. First, the supplier property tables end at 982 °C for k and 1093 °C for c_p , below the solidus [22]. The L/H branches are continuation scenarios, not measured liquid-state functions. Composition-based Ni-superalloy estimates explicitly include IN625 conductivity, but the underlying liquid-state data remain limited [26]. Such estimates provide candidates for a material-specific comparison; they do not select either continuation for this coupon. Second, the model excludes evaporation, melt flow and surface evolution. Its LL, HL, LH and HH surface peaks are approximately 2697, 3507, 2756 and 3721 °C. In particular, the held-conductivity peaks cannot be interpreted as credible physical-temperature predictions from this balance. The model nevertheless exposes how those assumptions affect the fitted geometry and rear thresholds.

Numerical checks support the larger responses without removing these physical limits. The main 34–43 μm length changes exceed the inspected baseline axial sensitivity, while the submicrometre interactions lack branch-specific three-dimensional refinement evidence. The analytical comparison, energy balance and solver correction test numerical execution; they do not validate liquid properties or a thermal trajectory. The experimental comparisons are also retrospective and combine length and cross-section quantities from separate observation populations, with descriptive uncertainty scales rather than a joint likelihood.

A discriminating thermal test would retain a defined source and observation mapping, reserve tracks or profiles before fitting, and compare an excluded temperature trajectory together with width and depth. Composition- and state-matched high-temperature property evidence would address which continuation is physically appropriate; a matched thermal trajectory would address whether the calibrated field is credible. Until those tests are available, the result is specific but useful: one geometric fit leaves distinct questions about fusion shape, tail position, phase separation and material cooling time, and the fixed-source continuation contrast reveals which of those questions the assumed properties influence.

5. Conclusions

Geometric calibration provides a useful constraint on thermal modeling, but its solidification meaning depends on the properties and observables used to interpret the fitted field. For the present IN625 enthalpy model, the source factor fitted to a surface length leaves a width excess of $19.54\ \mu\text{m}$ and a depth deficit of $4.88\ \mu\text{m}$. The faster fixed-source condition remains wide and shallow. The excluded shape quantities therefore supply information that the length fit does not capture.

The fixed-source property contrast separates opposing responses that a joint continuation change conceals. Conductivity holding lengthens the pool by $43.21\ \mu\text{m}$, sensible heat-capacity holding shortens it by $6.17\ \mu\text{m}$, and their interaction gives a joint extension of $34.28\ \mu\text{m}$. Rear solidus displacement accounts for 98.3% of the conductivity-induced extension, while the rear phase separation changes much less. The dominant response is displacement of the thermal tail, with opposing depth effects that do not resolve the geometric discrepancy.

Spatial phase separation and material cooling time must also be distinguished. Rear spans near $80\ \mu\text{m}$ at 0.8 and 1.2 m/s correspond to passage times of 100.33 and $67.33\ \mu\text{s}$ and a 49.02% difference in the interval-mean cooling rate. These are transformations of the modeled centerline field. Together, the results show why thermal interpretation after a geometric fit requires separate assessment of fusion shape, property response and material time. Aligned source and observation definitions, credible high-temperature data and an independent thermal profile are needed to test that interpretation physically.

Data and code availability

The LaTeX manuscript, vector figures, extracted tables and evidence inspection are available in the [manuscript package](#). The original case and numerical implementation are referenced through [SimAgent](#). Experimental sources remain the cited NIST challenge and Lane study. Publisher full-text copies used for reading are not redistributed. The package README distinguishes extracted tables and figures from large retained fields and identifies the corresponding original case locators.

Declaration of generative AI use in manuscript preparation

OpenAI Codex was used to assist literature and evidence organization, manuscript drafting and automated pre-review. Figures were generated from numerical results by deterministic plotting.

Appendix A. Material data and numerical verification

Appendix A.1. Material-property inputs

The continuation slopes from the final tabulated pairs are $0.0216216\ \text{W m}^{-1}\ \text{K}^{-2}$ for conductivity and $0.225225\ \text{J kg}^{-1}\ \text{K}^{-2}$ for sensible specific heat. The enthalpy integral is evaluated piecewise analytically and inverted using the corresponding property continuation.

Table A.4: Thermophysical-property inputs from the general IN625 bulletin [22]. The two temperature columns describe different tables. Conductivity and sensible specific heat are interpolated independently.

T_k (°C)	k (W m ⁻¹ K ⁻¹)	T_c (°C)	c_p (J kg ⁻¹ K ⁻¹)
-157	7.2	-18	402
-129	7.5	21	410
-73	8.4	93	427
-18	9.2	204	456
21	9.8	316	481
38	10.1	427	511
93	10.8	538	536
204	12.5	649	565
316	14.1	760	590
427	15.7	871	620
538	17.5	982	645
649	19.0	1093	670
760	20.8		
871	22.8		
982	25.2		

Appendix A.2. Spatial discretization and observation recovery

The domain is $\xi \in [-1, 0.35]$ mm, $y \in [0, 0.36]$ mm and $z \in [-0.30, 0]$ mm. A graded orthogonal mesh contains 902664 cells. Near-source cell widths are $4 \times 3 \times 1$ μm within $\xi \in [-0.48, 0.16]$ mm, $y \in [0, 0.12]$ mm and $z \in [-0.06, 0]$ mm. Outer widths increase by a factor of 1.22 up to 25 μm . The half-domain mesh has $199 \times 56 \times 81$ cells. The solved problem is steady; retained warm fields provide nonlinear initial guesses rather than a transient thermal initial condition.

The Gaussian source is integrated rectangle-wise using error functions to supply each top-face power, which is deposited in the adjacent top control volume. Its full-plane integral is ηP and the transverse half-domain receives $\eta P/2$.

Surface recovery satisfies

$$K(T_{\text{surf}}) - K(T_{\text{cell}}) = q \Delta z/2, \quad K'(T) = k(T), \quad (\text{A.1})$$

and the symmetry-plane value uses the even quadratic extension $T(0) = T(y_1) + [T(y_1) - T(y_2)]y_1^2/(y_2^2 - y_1^2)$ from the first two positive- y cell centers. The recovered surface temperatures and symmetry extension are applied before the threshold crossings are extracted.

The observation pitch is 0.5 μm ; it samples the interpolated geometry rather than changing the heat-field resolution.

Appendix A.3. Scaled enthalpy diffusion and diagnostic definitions

The scaled unknown is $U = H/C_0$, with $C_0 = \rho(402 \text{ J kg}^{-1} \text{ K}^{-1}) = 3.39288 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$. On an internal face,

$$D_f = \frac{k_f}{C_0} \frac{\Delta T}{\Delta U}, \quad (\text{A.2})$$

where k_f is the harmonic face conductivity and D_f has units of $\text{m}^2 \text{ s}^{-1}$. For $|\Delta U| \leq 10^{-8} \text{ K}$, the derivative limit is $\Delta T/\Delta U = C_0/[\rho(c_p + \mathcal{L}f')]$ at the first adjacent cell. This recovers conductive flux while conserving enthalpy; U is a scaled enthalpy, not temperature.

The retained face transport diagnostic is

$$|\text{Pe}_f| = \frac{|\mathbf{u} \cdot \mathbf{n}_f| d_f}{D_f}, \quad (\text{A.3})$$

where d_f is the adjacent cell-center separation. Unweighted medians are taken over internal axial faces adjacent to at least one cell with $T_s \leq T < T_l$. The diagnostic compares coordinate transport with diffusion in the discretized field; its selected face population can change between solved branches.

The discrete rear hot-region budget selects cells with $T \geq T_s$ and $\xi < 0$. Net outward diffusive power is the sum of signed conductive face powers on the boundary of this selected cell set; internal face contributions cancel. The region is selected separately in each field, so its boundary and sampled faces change with the continuation. This accounting diagnostic is interpreted alongside the matched-coordinate centerline thresholds rather than as a fixed-region causal contrast.

Appendix A.4. Scalar calibration and solution acceptance

A safeguarded secant solves $L_B(\eta) - 359 \mu\text{m} = 0$ in the initial bracket $[0.2, 0.4]$, with proposals restricted to the inner 90% of the current bracket and an accepted absolute length residual below $0.5 \mu\text{m}$. It yields $\eta = 0.2890548519203547$, with a residual of $+0.158 \mu\text{m}$. The calibration record below gives trial values and the local sensitivity conversion. A and C retain the final value, the same beam and phase law, and the same continuation rules.

Nonlinear acceptance requires the sum of absolute assembled cell residuals, normalized by absorbed half-power, to be below 2×10^{-5} , the signed global balance magnitude below the same scale, and a further frozen-coefficient solve to change temperature by less than 0.002°C . Local residual and global balance are retained separately because opposing local errors can cancel in a global sum.

The six production solutions have finite temperatures, bounded phase fractions, nontruncated extracted geometry and the prescribed integrated source power. Normalized absolute residuals are 6.90×10^{-8} – 1.31×10^{-7} , signed global balance magnitudes are below 6.1×10^{-11} , and full-solve temperature corrections are below $2.7 \times 10^{-5}^\circ\text{C}$.

Appendix A.5. Separate constant-property verification problem

The Gaussian verification uses $P = 179.2 \text{ W}$, $\eta = 0.2$, $v = 0.8 \text{ m/s}$ and $w = 85 \mu\text{m}$, with $\rho = 8440 \text{ kg m}^{-3}$, constant $c_p = 410.94444 \text{ J kg}^{-1} \text{ K}^{-1}$, constant $k = 9.870588 \text{ W m}^{-1} \text{ K}^{-1}$ and no latent heat. Thus $a_0 = k/(\rho c_p) = 2.845886 \times 10^{-6} \text{ m}^2/\text{s}$. Its finite-domain mesh contains 178992 cells ($113 \times 33 \times 48$), with near-source spacings of $8 \times 6 \times 2 \mu\text{m}$. Domain extents follow the production geometry, but the adiabatic transverse and bottom boundaries truncate the semi-infinite analytical referent.

A constant-property moving-Gaussian reference is constructed from the semi-infinite heat kernel, in the moving-source formulation [7]. With $Q = \eta P$, convolving the surface Gaussian with the Neumann kernel gives

$$T - T_0 = \frac{Q}{\rho c_p \pi \sqrt{\pi a_0}} \int_0^\infty \frac{\exp\left[-\frac{(\xi + v\tau)^2 + y^2}{w^2/2 + 4a_0\tau} - \frac{z^2}{4a_0\tau}\right]}{\sqrt{\tau}(w^2/2 + 4a_0\tau)} d\tau. \quad (\text{A.4})$$

The archived implementation removes the surface singularity by $\tau = (w/v)s^2$ and uses adaptive quadrature. Solidus geometry is extracted at 1290°C , with uniform-core face-average source recovery for the numerical surface. Width/depth use the monotone constant-property maximum-temperature envelope; they do not replace the general production passage observer. The associated recipes and archived reports are identified in the numerical reproduction notes. The quoted differences combine discretization, reconstruction and finite-domain effects, and do not bound nonlinear phase-change or physical-model error.

Appendix A.6. Calibration record and retained sensitivities

The calibration bracket $\eta \in [0.2, 0.4]$ has retained endpoint lengths of 234.57 and 511.68 μm . Subsequent retained trial pairs $(\eta, L/\mu\text{m})$ are (0.2898051, 359.96) and (0.2853148, 354.22), followed by the final (0.2890548519, 359.16). They establish a sampled local bracket, not global identifiability. The local secant sensitivity is approximately 1276.79 μm per unit η . Dividing B’s 21.26 μm expanded uncertainty by this slope gives a scale $\Delta\eta = 0.01665$; it is not a posterior interval and is not propagated into the geometry comparisons.

Table A.5: Retained calibrated axial sensitivities in μm , calculated as fine minus baseline. Only the near-source axial spacing changes from 4 to 2 μm ; transverse and depth spacings remain 3 and 1 μm . These contrasts do not bound every discretization component.

Condition	ΔL	ΔW	ΔD
B	+0.361	−0.009	+0.041
C	+0.199	−0.029	+0.045

For uncertainty-scale reporting, $\delta/U = (Q_{\text{model}} - \overline{Q}_{\text{source}})/U_{\text{Lane}}$. The separate U values are not specimen standard deviations. Imaging and microscopy have different operators and aggregation, so the geometry triplet is not fitted to a joint likelihood. All finite contrasts use full-precision outputs before rounding for display.

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